

303×317 interaction maps, scenario C ($\rho = 2.0$ USD/kg) — $I = f_{\text{pair}} - f_{303} - f_{317}$; $I > 0$ for Δ metrics = sub-additive (substitutes). Grey cells: $|I| < \sqrt{3} \times \text{single-measurement floor}$ (I sums three independent noisy measurements). floors: cost 0.5M USD, curt 5 GWh, shed 3 GWh, starts = quadratic residual RMSE 48 (working floor — no stated floor)

